Graphs 5 — Inequalities on graphs · Perpendicular lines · Mid-points · Distances

Squared paper needed. Shade the regions *not* wanted, as in the exam. Marks in brackets. ★ = harder question.

1 Illustrate the inequality $-2 < x \leq 4$ on a graph, shading the unwanted regions. [2]

> Draw $x = -2$ as a **dotted** line (not included) and $x = 4$ as a **solid** line (included). Shade everything to the left of $x = -2$ and to the right of $x = 4$; the strip between them is the region.

2 **a** Draw axes from 0 to 10 and on them draw the lines $x = 5$, $y = 7$ and $2x + 3y = 24$. [8]

b Shade the regions not satisfied by $x \leq 5$, $y \leq 7$, $2x + 3y \leq 24$, $x \geq 0$, $y \geq 0$.

c Write down the point with integer coordinates in the unshaded region that is furthest from the origin.

> **a** $x = 5$ vertical, $y = 7$ horizontal; $2x + 3y = 24$ through $(0, 8)$ and $(12, 0)$ — use $(0, 8)$, $(3, 6)$, $(9, 2)$. All solid lines.
> **b** Keep the region below $y = 7$, left of $x = 5$, below the sloping line, in the first quadrant.
> **c** Candidates near the far corner: $(1, 7)$: $2 + 21 = 23 \leq 24$ ✓, distance $\sqrt{50} = 7.07$; $(3, 6)$: on the line, distance $\sqrt{45} = 6.71$; $(5, 4)$: $\sqrt{41} = 6.40$. Furthest: $(1, 7)$. ($(2, 7)$ gives $25 > 24$ — outside.)

3 A is $(2, -3)$ and B is $(6, 5)$. [7]

a Find the length of AB.

b Find the mid-point of AB.

c Find the equation of the perpendicular bisector of AB.

> **a** $\sqrt{(6-2)^2 + (5-(-3))^2} = \sqrt{16 + 64} = \sqrt{80} = 8.94$ (3 s.f.).
> **b** $\left(\frac{2+6}{2}, \frac{-3+5}{2}\right) = (4, 1)$.
> **c** Gradient of $AB = \frac{8}{4} = 2$, so the perpendicular gradient is $-\frac{1}{2}$. Through $(4, 1)$: $1 = -\frac{1}{2}(4) + c \Rightarrow c = 3$. Equation: $y = -\frac{1}{2}x + 3$.

4 A is $(1, -4)$ and B is $(7, 4)$. AB is a diameter of a circle with centre C. [8]

- **a** Find the coordinates of C.
- **b** Find the radius of the circle.
- **c** Show that the point $D(8, 3)$ lies on the circle.
- **d** A circle theorem says that AD is perpendicular to BD. Show that this is true.

> **a** Mid-point of AB: $(4, 0)$.
> **b** $AB = \sqrt{36 + 64} = 10$, so the radius is 5.
> **c** $CD = \sqrt{(8-4)^2 + (3-0)^2} = \sqrt{25} = 5$ = the radius, so D is on the circle.
> **d** Gradient $AD = \dfrac{3-(-4)}{8-1} = 1$; gradient $BD = \dfrac{3-4}{8-7} = -1$. Product $= -1$, so the lines are perpendicular — the angle in a semicircle is $90°$.

5 Find the equation of the line perpendicular to $y = 3x - 2$ that passes through $(6, 1)$. [3]

> Perpendicular gradient $-\frac{1}{3}$. $1 = -\frac{1}{3}(6) + c \Rightarrow c = 3$. $y = -\frac{1}{3}x + 3$.

Total: 28 marks

Check: solid line for $\leq$ / $\geq$, dotted for $<$ / $>$ · test a point to decide which side · perpendicular gradients multiply to -1 · mid-point = mean of the coordinates · distance = Pythagoras on the coordinate differences.